HoleKing

How to run it

Hole–electron and charge-transfer analysis of sTDA-xTB excited states

HoleKing takes a molecule (an `.xyz` file), computes its excited states with sTDA-xTB or sTD-DFT-xTB, and tells you where the hole and the electron go in each state. You get one CSV file with one row per state. This guide takes you from a fresh Linux machine to that CSV.

Set it up (once)

Every command below goes in a Linux terminal. You need Python 3.9 or newer and git.

Step 1 Get xtb4stda and stda

HoleKing doesn't compute the excited states itself; these two programs from the Grimme group do. Download both programs and their parameter files into a folder called `xtb4stda` in your home:

```
$ cd ~
$ git clone https://github.com/grimme-lab/xtb4stda.git   # parameter files
$ cd xtb4stda
$ wget https://github.com/grimme-lab/xtb4stda/releases/download/v1.0/xtb4stda
$ wget https://github.com/grimme-lab/xtb4stda/releases/download/v1.0/stda_v1.6.1
$ chmod +x xtb4stda stda_v1.6.1                          # make them executable
```

If they are already installed on your machine or cluster, skip this step and use their paths in Step 2.

Step 2 Tell HoleKing where they are

HoleKing finds the programs through three environment variables. Adding them to `~/.bashrc` means you only do this once:

```
$ echo 'export XTB4STDA_BIN=~/xtb4stda/xtb4stda' >> ~/.bashrc
$ echo 'export STDA_BIN=~/xtb4stda/stda_v1.6.1' >> ~/.bashrc
$ echo 'export XTB4STDAHOME=~/xtb4stda' >> ~/.bashrc
$ source ~/.bashrc
```

`XTB4STDAHOME` is the folder with the `.param_stda1.xtb`, `.param_stda2.xtb` and `.param_gbsa_<solvent>` files.

Step 3 Download HoleKing

```
$ cd ~
$ git clone https://github.com/rafaelfv27/ElectronHoleKing.git
$ cd ElectronHoleKing
```

Step 4 Create a Python environment

This is a private folder for HoleKing's Python packages, so they don't clash with the rest of your system. Many Linux systems refuse `pip install` without one.

```
$ python3 -m venv ~/holeking-env
$ source ~/holeking-env/bin/activate
```

Your prompt now starts with `(holeking-env)`. **Every time you open a new terminal**, run the `source` line again before using HoleKing.

Step 5 Install HoleKing

Still inside the `ElectronHoleKing` folder:

```
$ pip install .        # HoleKing + numpy + pyscf
$ pip install rdkit    # optional
```

The first line installs HoleKing (the `.` means “the program in this folder”) together with numpy and pyscf, which it needs. The second is optional: rdkit is only used to split the molecule into fragments around a core (the BTD core option).

Step 6 Check that it works

```
$ holeking --help
```

You should see the list of commands. If you get `command not found`, the environment from Step 4 is not active: run `source ~/holeking-env/bin/activate`.

Run your first molecule

The repository comes with a small test molecule, benzothiadiazole. Copy it to a new folder and type `holeking` there. It asks a few questions; for this test press **Enter** on every one to keep the value in brackets.

```
$ mkdir ~/holeking-test && cd ~/holeking-test
$ cp ~/ElectronHoleKing/tests/data/btd/geom.xyz btd.xyz
$ holeking
HoleKing: hole-electron analysis of sTDA-xTB excited states. Press Enter to accept
[defaults].

xyz file(s) [btd.xyz]:
solvent (none, water, toluene, thf, ...) [none]:
method: tda = sTDA-xTB, rpa = sTD-DFT-xTB [tda]:
number of excited states [10]:
molecular charge [0]:
fragments: 1 = BTD core, 2 = other core (SMARTS), 3 = fragment file, 4 = none [1]:
save results to [btd.csv]:

Same as: holeking run btd.xyz --nstates 10 --charge 0 --core btd --csv btd.csv
```

For your own molecules, do the same in the folder with your `.xyz` files.

What each question means

xyz file(s)	One or more geometries. Wildcards work: `*.xyz`.
solvent	`none`, or one of: acetone, acetonitrile, benzene, ch2cl2, chcl3, cs2, dmso, ether, water (h2o), methanol, thf, toluene.
method	`tda` (sTDA-xTB) is the fast default. `rpa` (sTD-DFT-xTB) is a bit slower.
excited states	How many states to compute, counted from the lowest.
charge	Total charge of the molecule.
fragments	How to split the molecule into donor/acceptor parts. See *Fragments* below.
save results to	Name of the CSV file.

With more than one xyz file it also asks how many molecules to run at the same time. If you type something HoleKing can't use, it says why and asks again. **Ctrl+C** quits.

What you get

For each molecule HoleKing makes a folder `<name>_holeking/` (here `btd_holeking/`) with the xtb and stda files and their logs, and writes the CSV you asked for. The first columns look like this (2,1,3-benzothiadiazole, gas phase):

```
state,energy_ev,wavelength_nm,f,first_bright,sr,d_index,t_index
1,4.5825,270.56,0.2645,1,0.8997,0.0453,-1.711
2,4.6532,266.45,0.2073,0,0.8986,0.8825,-0.738
3,5.0907,243.55,1e-06,0,0.4877,0.5357,-0.537
```

`energy_ev`, `wavelength_nm`	Excitation energy and the matching wavelength.
`f`	Oscillator strength: how bright the state is.
`first_bright`	1 for the lowest state with real intensity (f at least 5 % of the largest).
`near_degenerate_bright`	1 if another bright state lies within 0.1 eV.
`sr`	Hole–electron overlap. Close to 1: local excitation. Close to 0: charge transfer.
`d_index`	Distance between the hole and electron centres (Å).
`t_index`	Positive means hole and electron are well separated (CT).
`hole_fragN_*`, `ele_fragN_*`	Fraction of the hole and of the electron on fragment N (`mulliken` and `becke`).
`dpop_fragN_*`	Net electrons gained (+) or lost (−) by fragment N.
`ct_intrinsic_*`	Fraction of the excitation that is real charge transfer between fragments.

Fragments

Fragments let you ask *“does the electron move from the donor to the acceptor?”*

1 · BTD core

The benzothiadiazole core and its hydrogens become fragment 1. Every piece attached to it becomes its own fragment. Needs rdkit.

2 · Other core

Same idea, but you describe the core with a SMARTS pattern. It must match exactly once in the molecule.

3 · Fragment file

A text file, Multiwfn style: one fragment per line, atom numbers start at 1, ranges allowed. Every atom must be in exactly one fragment.

```
# frags.txt: three fragments
1-12
13-25,40
26-39
```

Many molecules, scripts and clusters

Interactive mode is for one-off runs. For scripts, use the one-line command it prints:

```
$ holeking run dad.xyz --solvent h2o --core btd --csv dad.csv
$ holeking run *.xyz --jobs 8 --core btd --csv all.csv     # 8 molecules at a time
$ holeking analyze dad_holeking/ --fragments frags.txt    # redo the analysis only
$ holeking run --help                                     # every option
```

With several molecules the CSV gets an extra `xyz` column. Keep `--jobs` at or below the number of CPUs; one thread per job was the fastest setting we measured. `analyze` reuses an existing folder, so you can try other fragments without running xtb and stda again.

When something goes wrong

Message	What to do
`xtb4stda parameter files not found`	`XTB4STDAHOME` points to the wrong folder, or the `.param_gbsa_<solvent>` file for your solvent is missing.
`No such file or directory: 'xtb4stda'`	Set `XTB4STDA_BIN` and `STDA_BIN` (Step 2).
`core pattern matched 0 times`	The molecule has no BTD core, or your SMARTS is wrong. Pick another fragment option.
`core pattern matched 2 times`	The core appears twice. Use a fragment file instead.
`fragments must cover each of the N atoms exactly once`	An atom is missing or repeated in your fragment file.
`... was not written; see the logs`	xtb or stda failed. Read `xtb4stda.out` and `stda.out` in the `_holeking` folder.

Results agree with Multiwfn (Sr within 0.001, D within 0.03 Å), and every oscillator strength is recomputed and checked against stda. Code and tests: github.com/rafaelfv27/ElectronHoleKing. MIT license.